Lab7-Digital Forensics Laboratory Report

Monday, July 20, 2026 7:44 PM

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Institution: Humber College
Date: July 20, 2026
Subject: Computer Forensics -Data Acquisition

1. Objective

The primary objective of this laboratory exercise was to develop hands-on competencies in forensic data acquisition. This included preparing target storage media in a Linux environment and securely acquiring disk images using built-in command-line utilities while adhering to forensic best practices.

2. Environment and Infrastructure

The practical exercises were conducted within a virtualized forensic laboratory network. The primary infrastructure utilized included:

- **PLABKSRV01:** A Kali Linux Workstation utilized as the primary forensic analysis machine.
- **PLABWIN10:** A Windows 10 Standalone Workstation.
- **PLABDEFT01:** An additional Linux Workstation.

3. Procedures Performed

3.1. Preparing a Target Drive for Acquisition in Linux (Exercise 3-1)

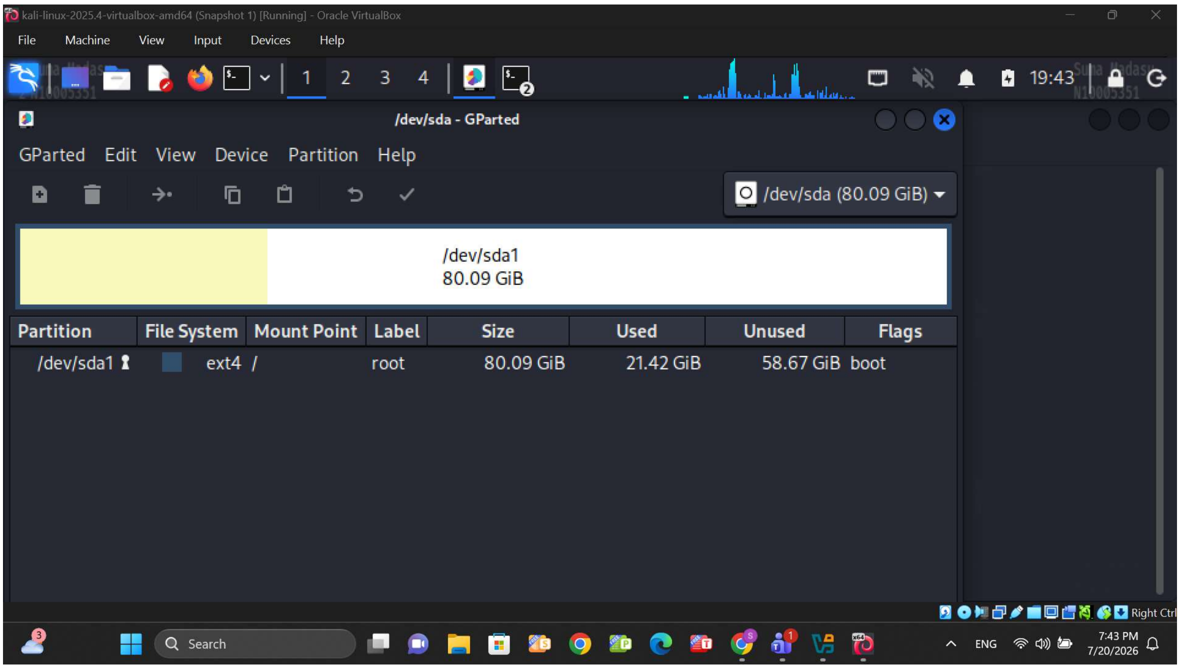
This phase established a forensically sound target drive formatted to Microsoft FAT32, eliminating the need to transition to a Windows operating system during prep.

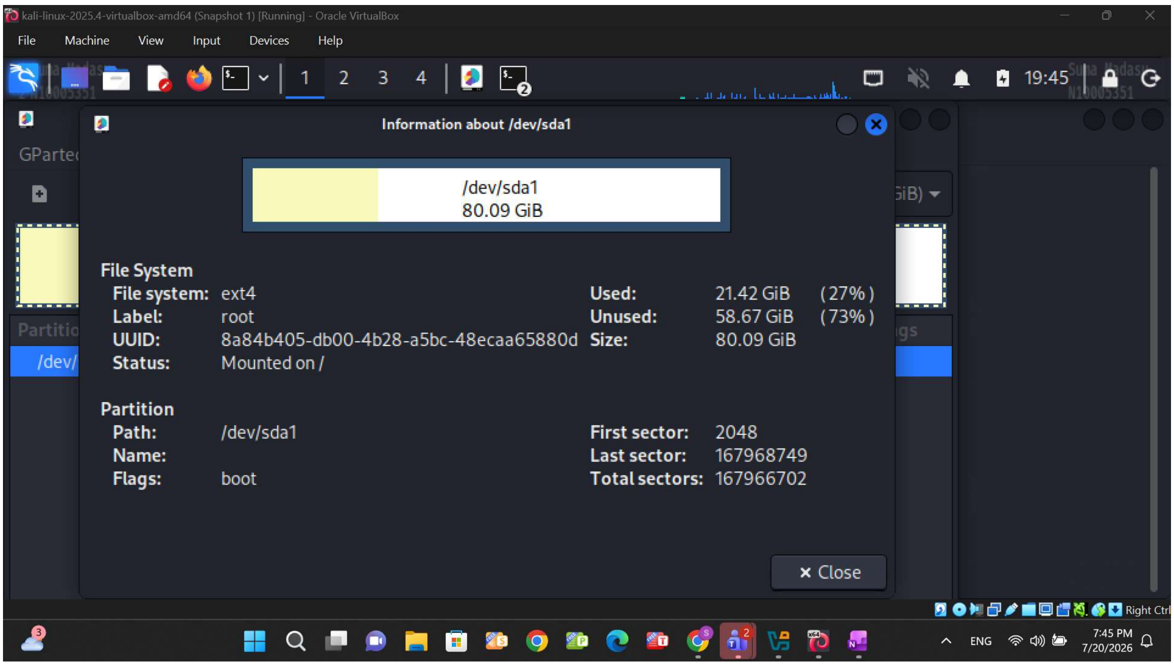
- **Disk Partitioning:** Accessed the Kali Linux workstation and utilized the gparted graphical utility to unmount and delete the existing partitions on the /dev/sdb drive. A new 4096 MiB primary partition was generated, formatted to the FAT32 file system, and labeled "Windows".
- **Volume Identification:** Executed the fdisk -l command in the terminal to map the partitioning details, verifying the block dimensions and confirming the W95 FAT32 system identifier.
- **Persistent Mounting Configuration:** Established a dedicated mount point directory via the mkdir /mnt/sdb1 command. The /etc/fstab configuration file was then modified using the leafpad text editor to append the new device path (/dev/sdb1 /mnt/sdb1 vfat defaults 0 0), ensuring the file system mounts correctly.

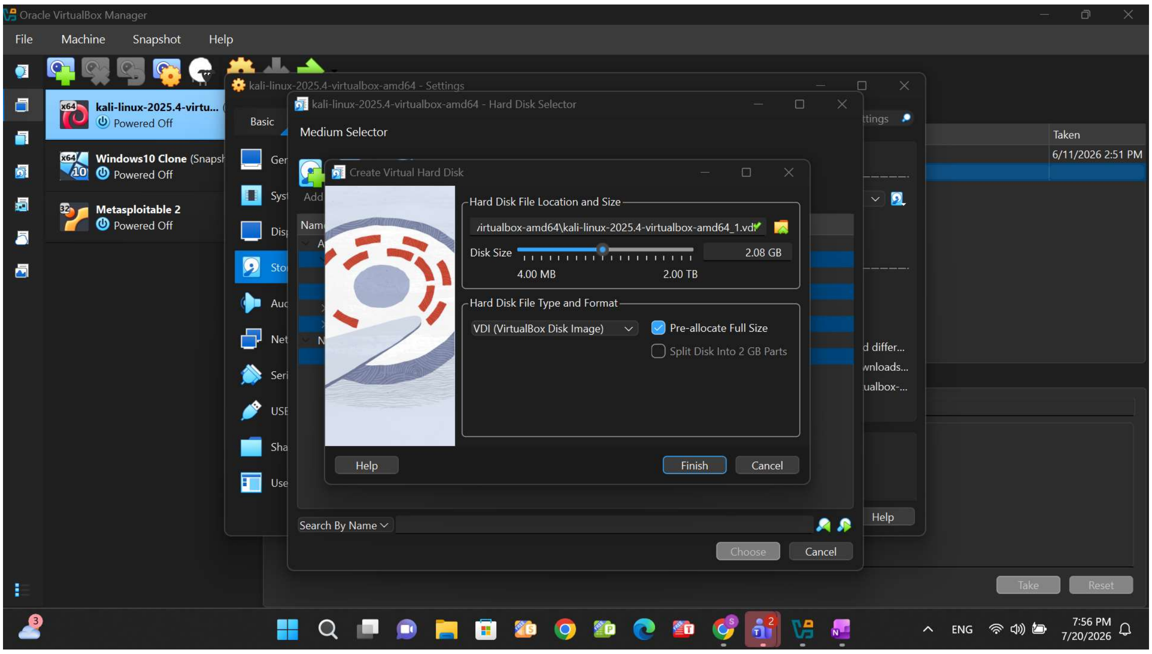
3.2. Acquiring Data with dd in Linux (Exercise 3-2)

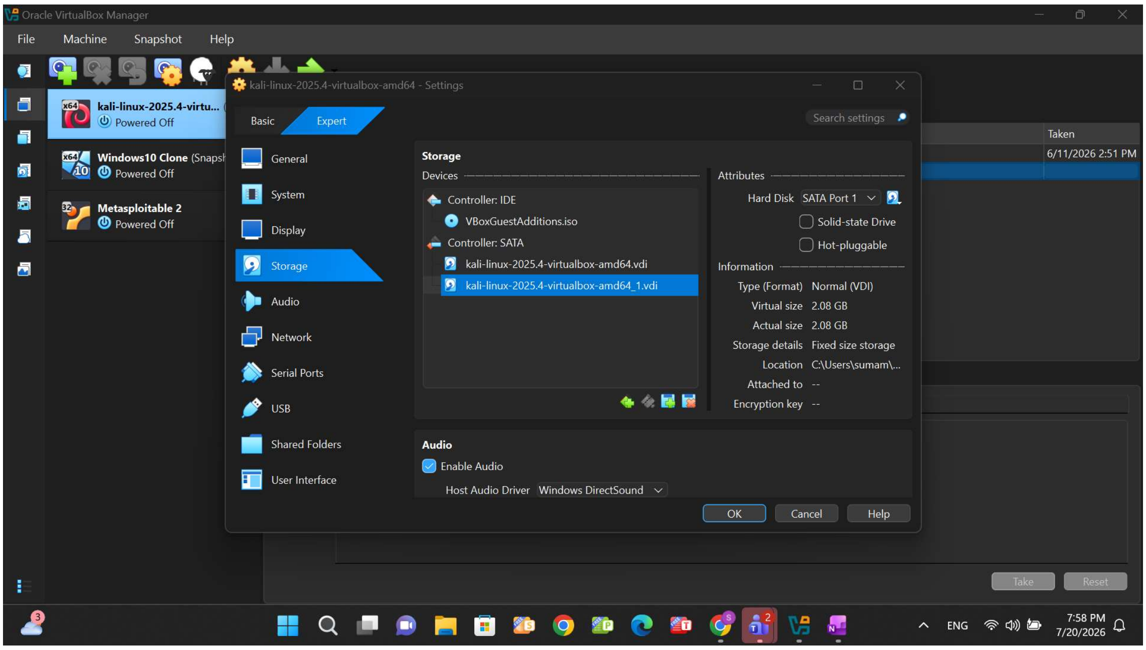
This phase focused on executing a raw format image of an NTFS disk onto the newly prepared FAT32 target disk.

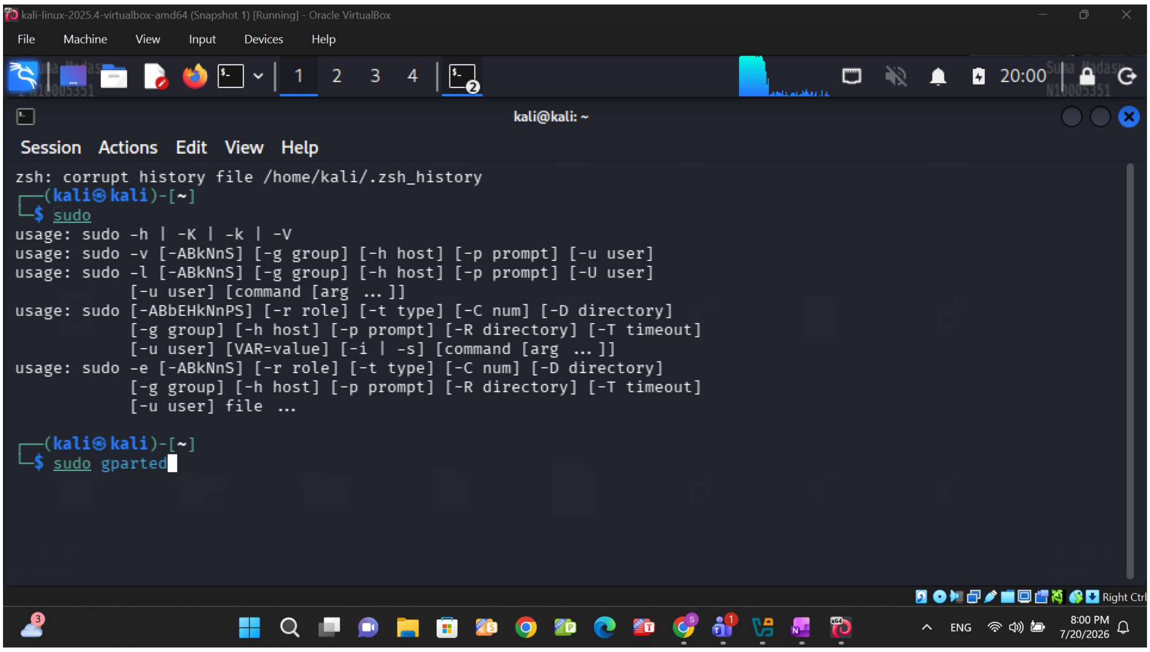
- **Target Directory Initialization:** Created a specific directory named case01 within the mounted volume to securely house the extracted evidence files.
- **Image Extraction Pipeline:** Executed a comprehensive dd command pipeline to duplicate the disk: dd if=/dev/sdb1 conv=sync,noerror bs=64k | gzip -c | split -b 2000m /system_drive_backup.img.gz.
- **File Size Management:** The split parameter was intentionally deployed to divide the resulting compressed image into 2 GB (2000m) segmented volumes. This ensures full compliance with the maximum file size limits inherent to the FAT32 file system architecture.

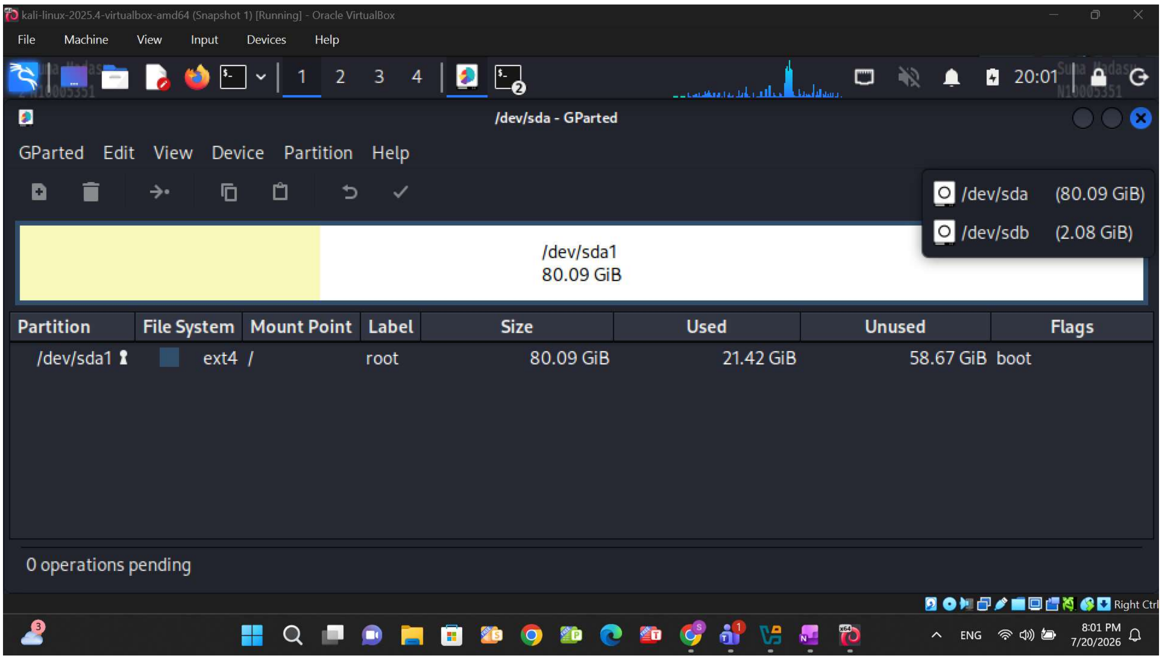


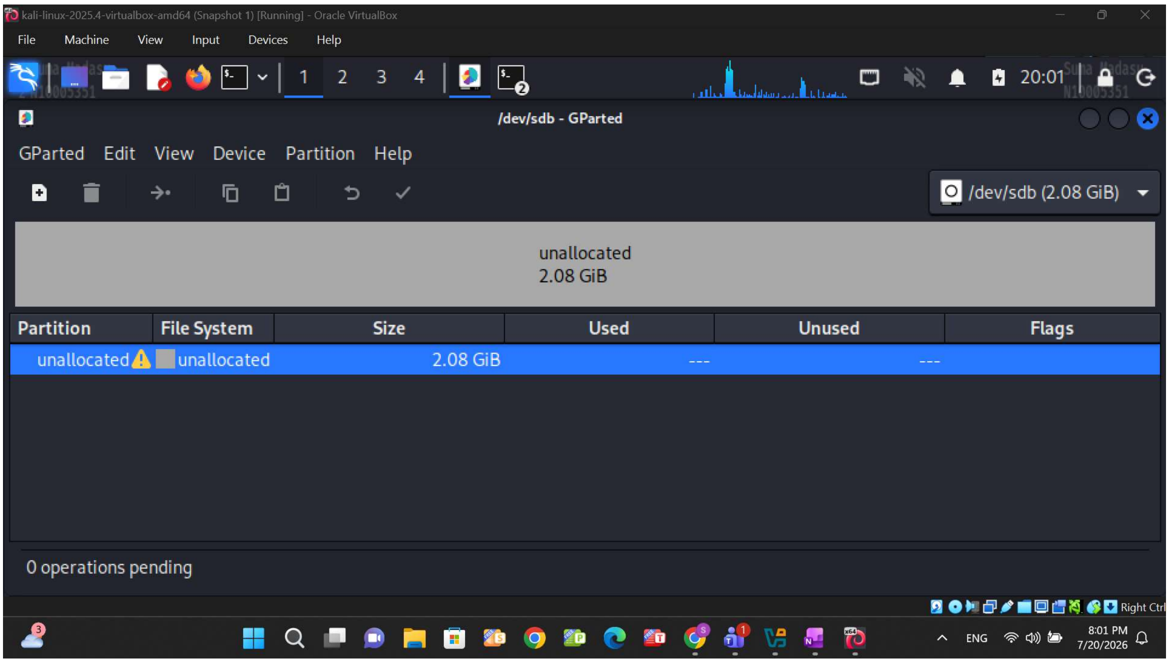


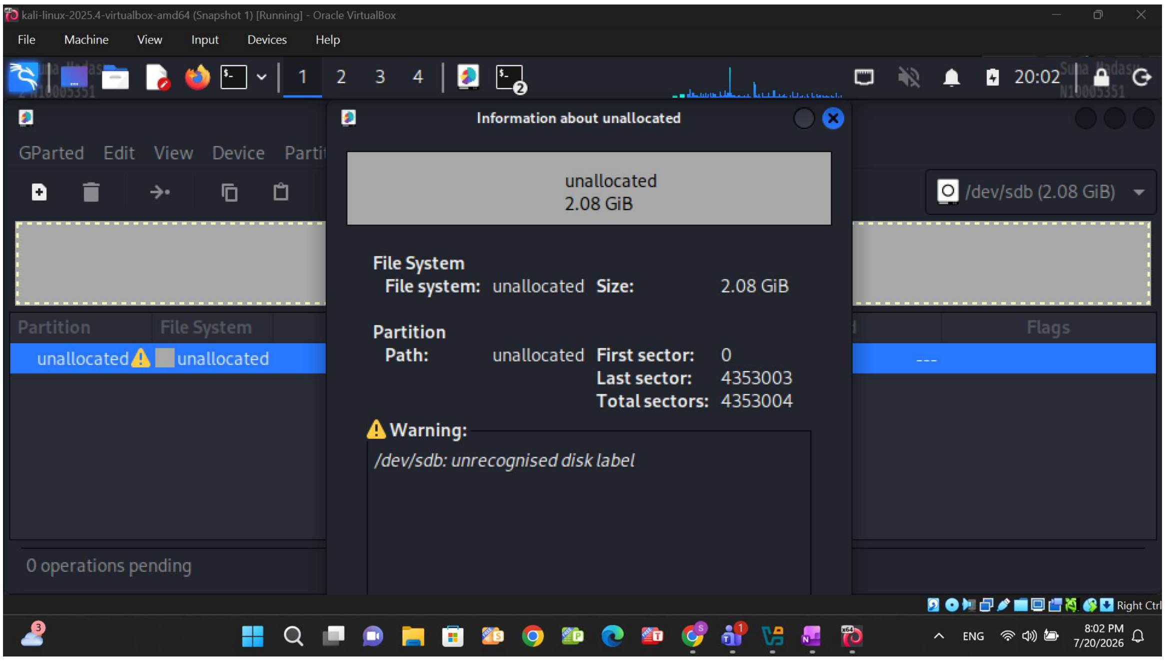


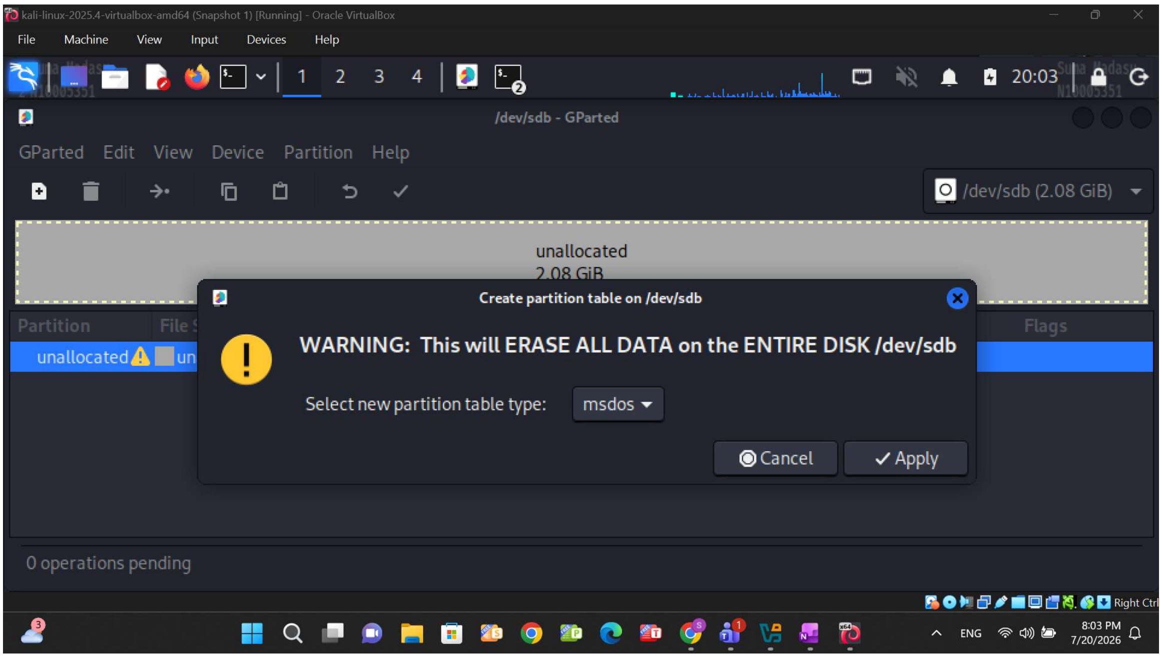


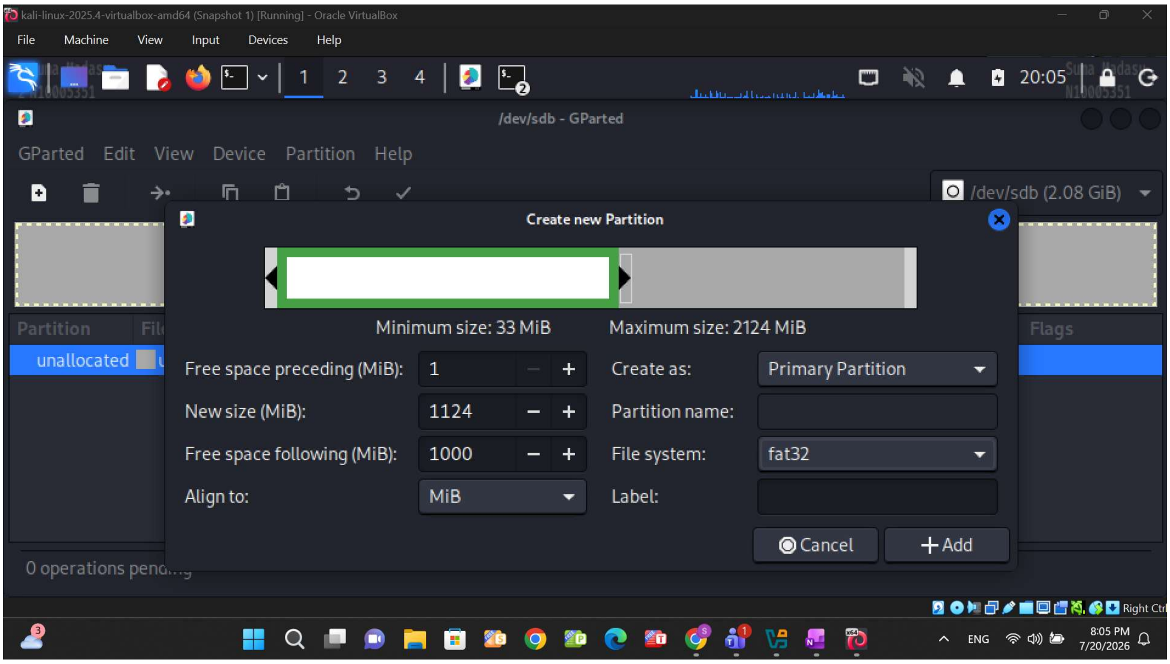

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ sudo
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
[-u user] [command [arg ...]]
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
[-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
[-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
[-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
[-u user] file ...
┌──(kali㉿kali)-[~]
└─$ sudo gparted
8:00 PM
7/20/2026

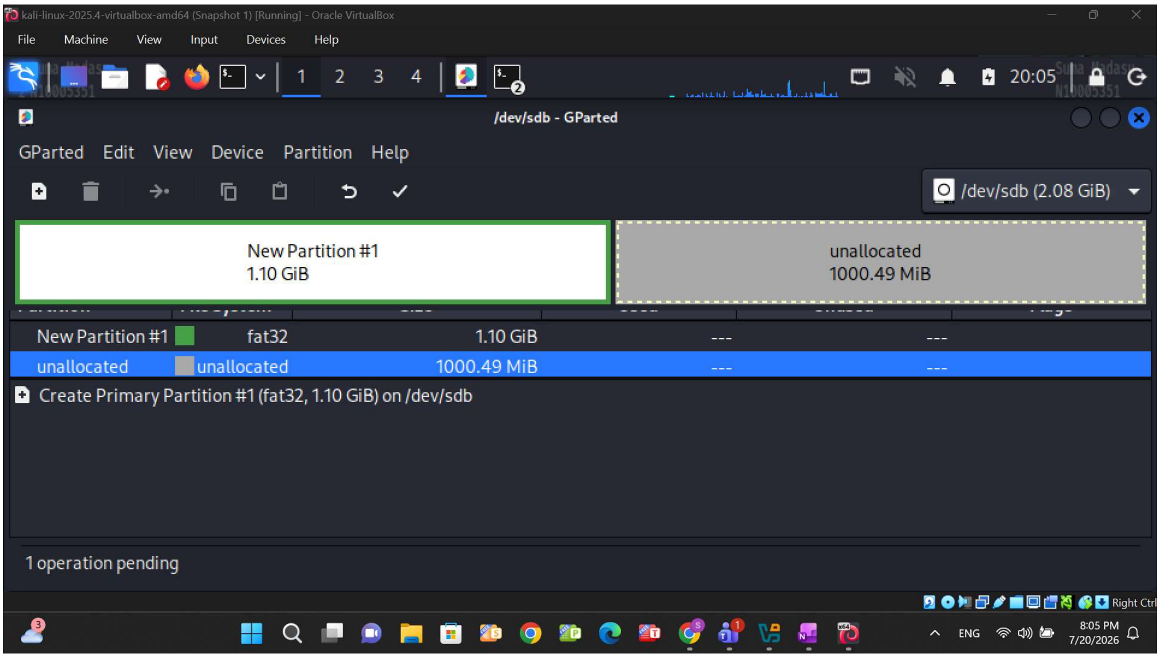

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
/dev/sda - GParted
GParted Edit View Device Partition Help
/dev/sda (80.09 GiB)
/dev/sdb (2.08 GiB)
/dev/sda1
80.09 GiB
Partition File System Mount Point Label Size Used Unused Flags
/dev/sda1 ext4 / root 80.09 GiB 21.42 GiB 58.67 GiB boot
0 operations pending
8:01 PM
7/20/2026

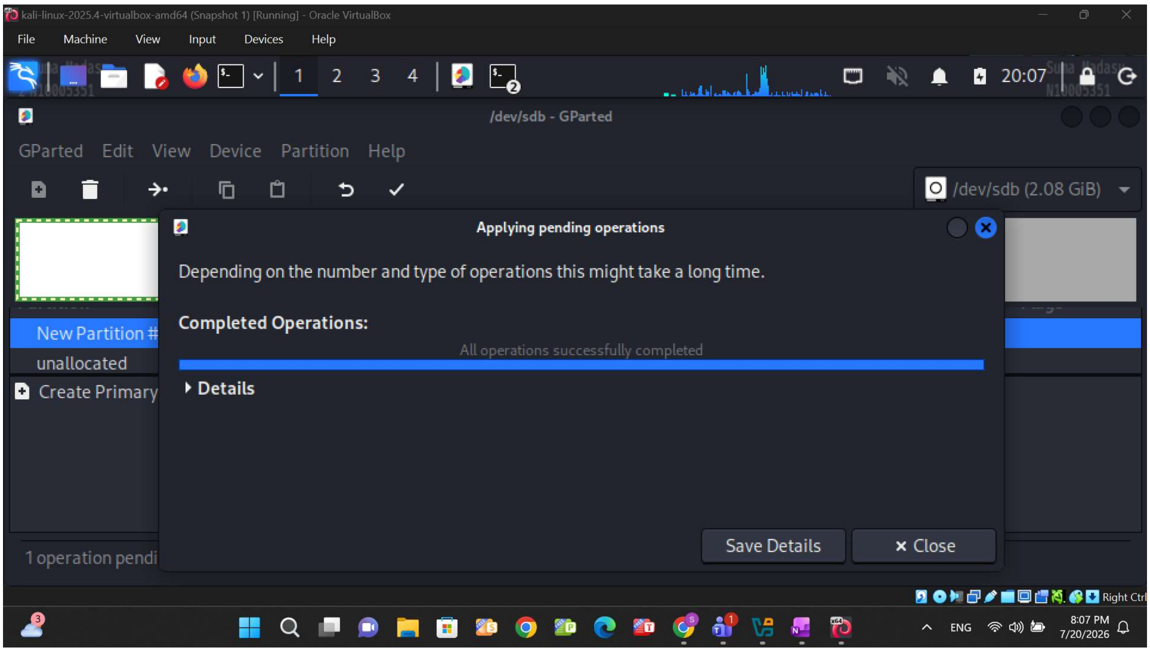

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
/dev/sdb - GParted
GParted Edit View Device Partition Help
/dev/sdb (2.08 GiB)
unallocated
2.08 GiB
Partition File System Size Used Unused Flags
unallocated unallocated 2.08 GiB --- ---
0 operations pending
8:01 PM
7/20/2026

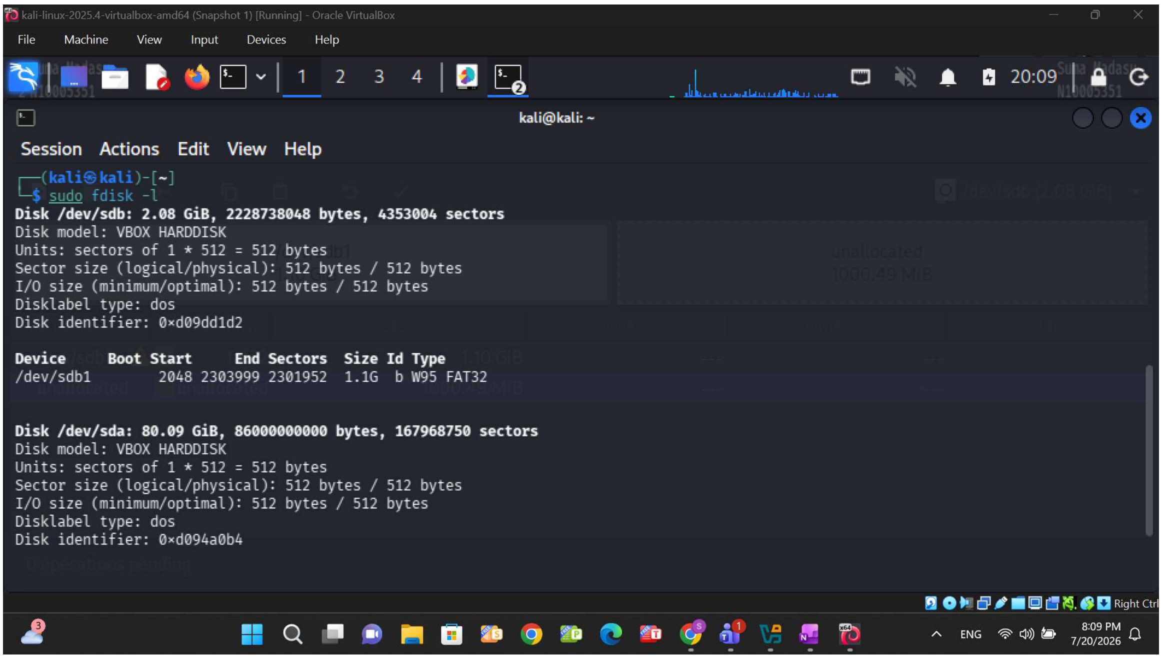

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
Information about unallocated
unallocated
2.08 GiB
File System
File system: unallocated Size: 2.08 GiB
Partition
Path: unallocated First sector: 0
Last sector: 4353003
Total sectors: 4353004
Warning:
/dev/sdb: unrecognised disk label
/dev/sdb (2.08 GiB)
0 operations pending

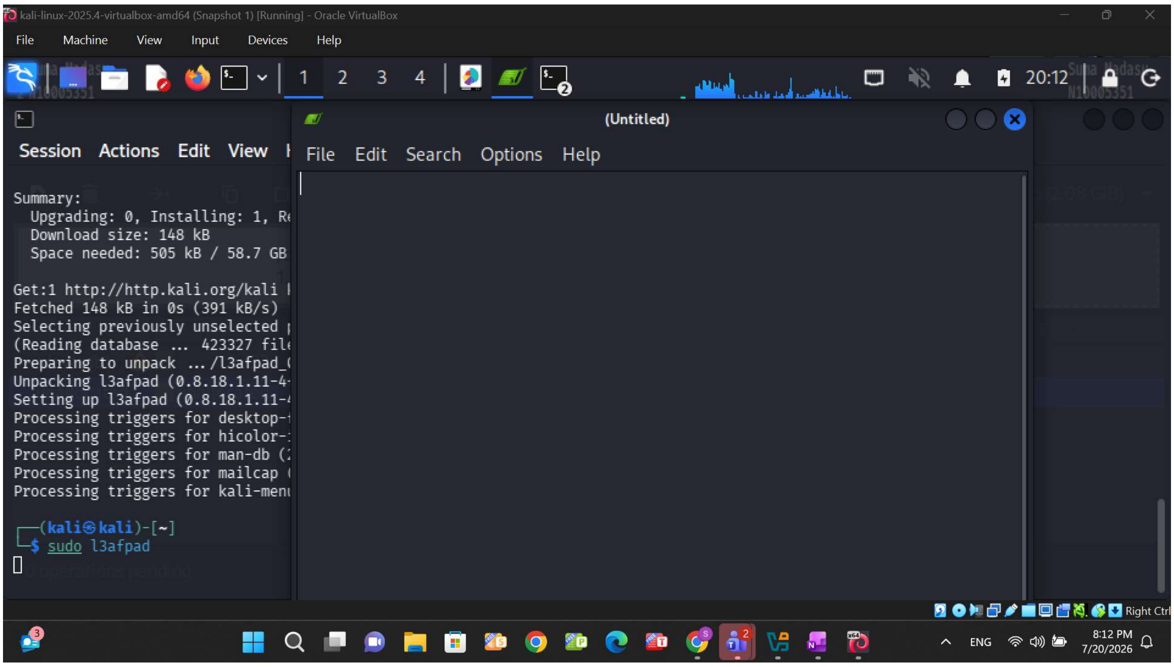

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
/dev/sdb - GParted
GParted Edit View Device Partition Help
/dev/sdb (2.08 GiB)
Create partition table on /dev/sdb
WARNING: This will ERASE ALL DATA on the ENTIRE DISK /dev/sdb
Select new partition table type: msdos
Cancel
Apply
0 operations pending

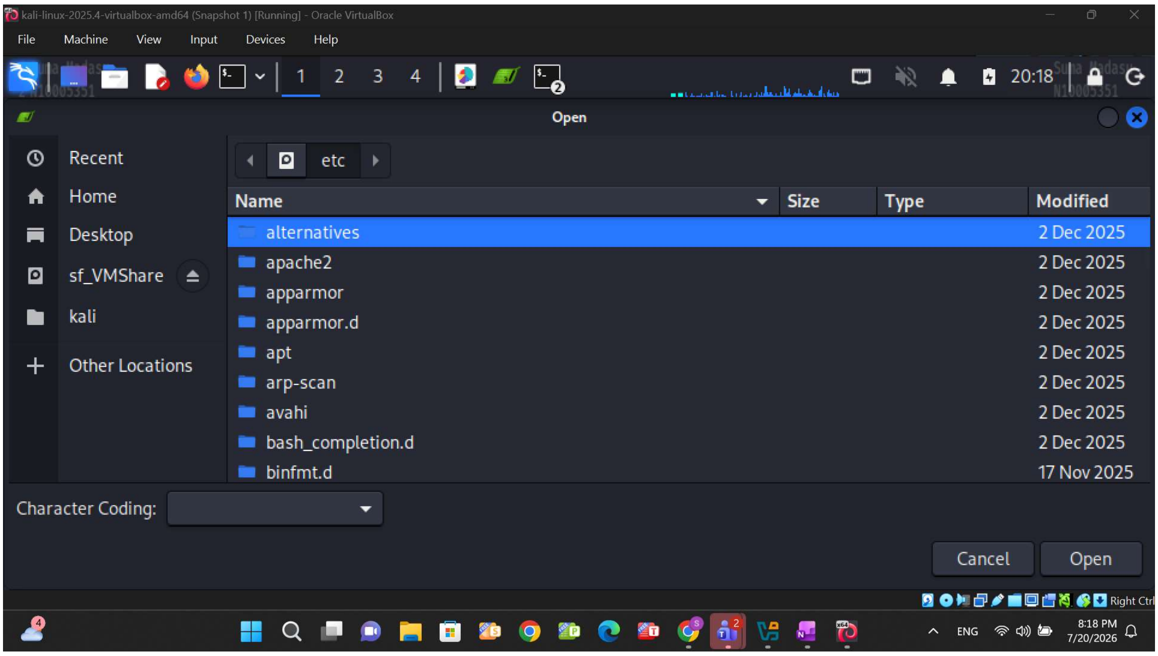

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
20:05
/dev/sdb - GParted
GParted Edit View Device Partition Help
/dev/sdb (2.08 GiB)
Create new Partition
Minimum size: 33 MiB
Maximum size: 2124 MiB
Free space preceding (MiB): 1
New size (MiB): 1124
Free space following (MiB): 1000
Align to: MiB
Create as: Primary Partition
Partition name:
File system: fat32
Label:
Cancel
Add
Partition
unallocated
Flags
0 operations pending
8:05 PM
7/20/2026

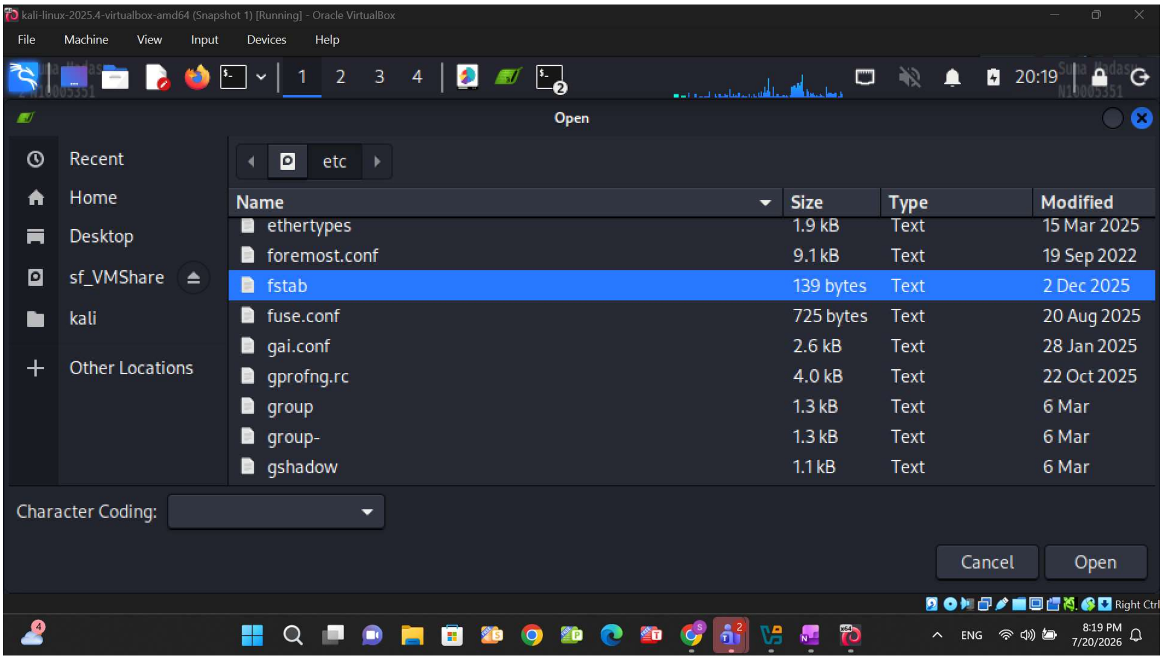

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
20:05
/dev/sdb - GParted
GParted Edit View Device Partition Help
/dev/sdb (2.08 GiB)
New Partition #1
1.10 GiB
unallocated
1000.49 MiB
New Partition #1 fat32 1.10 GiB --- ---
unallocated unallocated 1000.49 MiB --- ---
Create Primary Partition #1 (fat32, 1.10 GiB) on /dev/sdb
1 operation pending
8:05 PM
7/20/2026

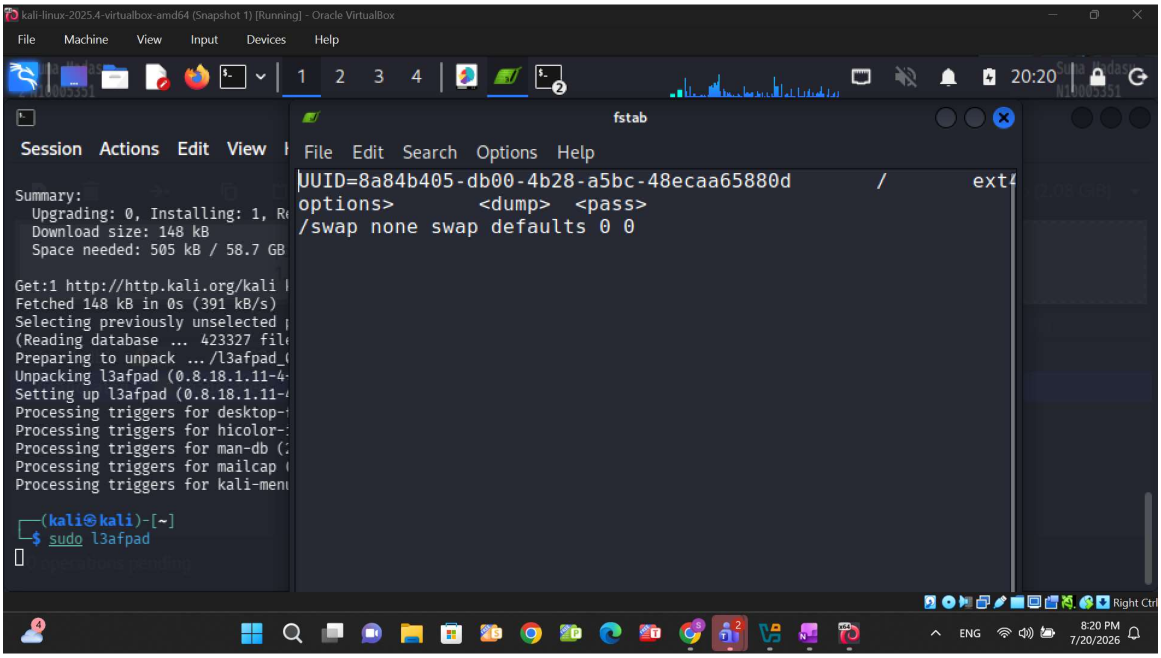

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
20:07
/dev/sdb - GParted
GParted Edit View Device Partition Help
/dev/sdb (2.08 GiB)
Applying pending operations
Depending on the number and type of operations this might take a long time.
Completed Operations:
All operations successfully completed
Details
Save Details
Close
New Partition #
unallocated
Create Primary
1 operation pendi
8:07 PM
7/20/2026

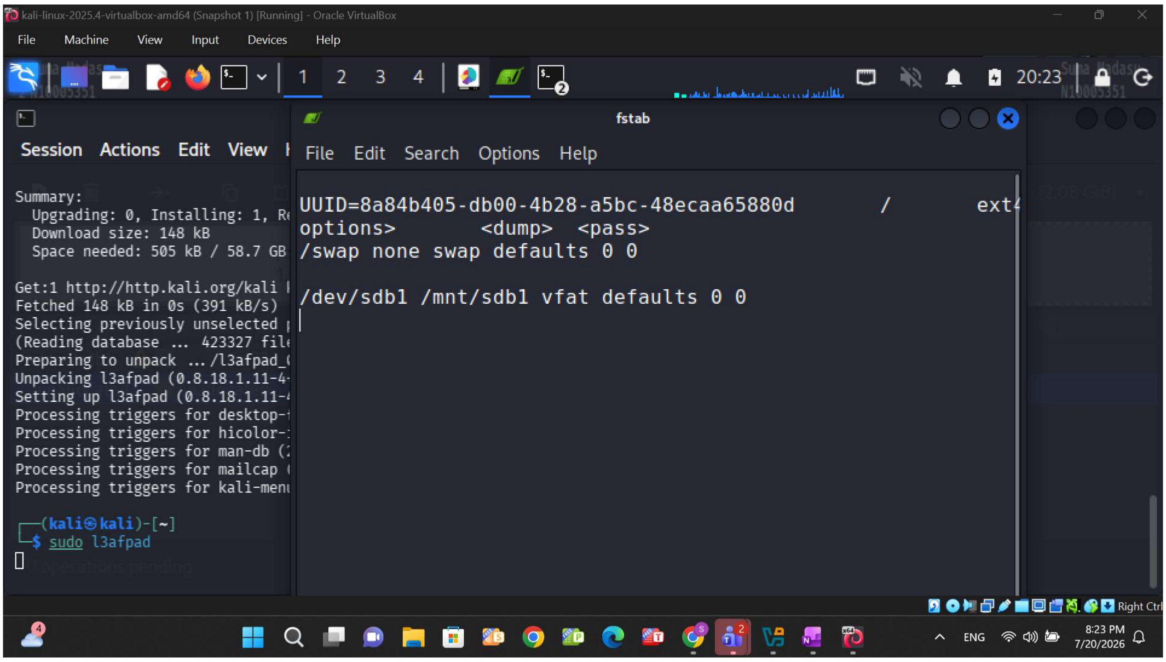

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
20:09
kali@kali: ~
Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo fdisk -l
Disk /dev/sdb: 2.08 GiB, 2228738048 bytes, 4353004 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0×d09dd1d2
Device Boot Start End Sectors Size Id Type
/dev/sdb1 2048 2303999 2301952 1.1G b W95 FAT32
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0×d094a0b4
Right Ctrl
ENG
8:09 PM
7/20/2026

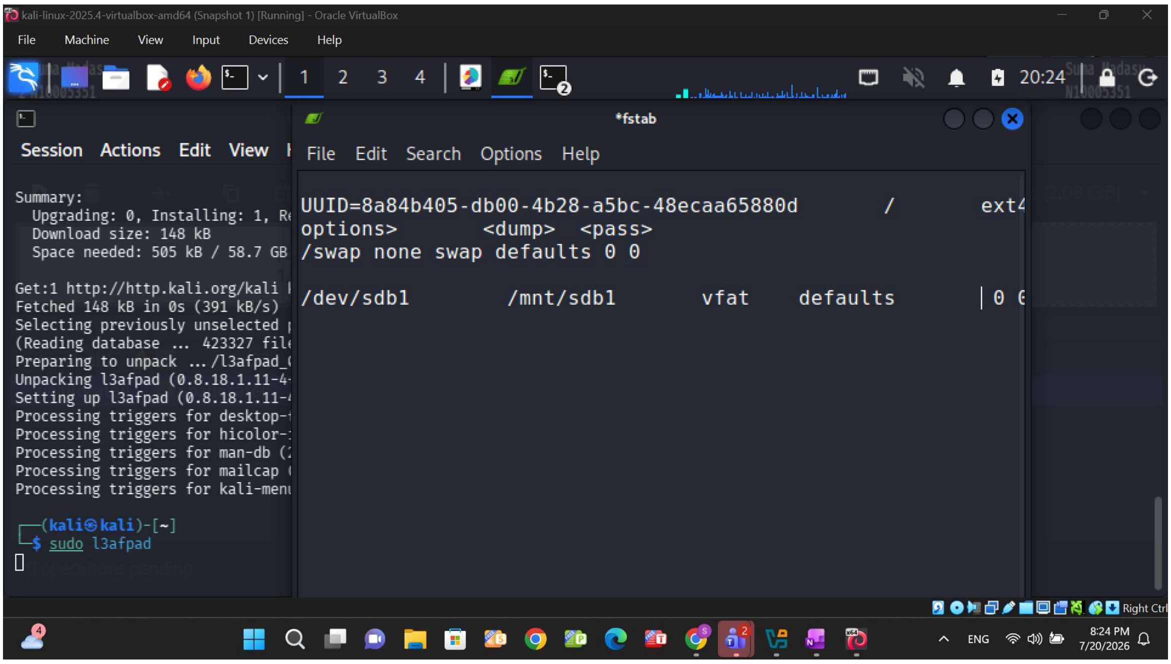

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
20:12
Session Actions Edit View
Summary:
Upgrading: 0, Installing: 1, Re
Download size: 148 kB
Space needed: 505 kB / 58.7 GB
Get:1 http://http.kali.org/kali
Fetched 148 kB in 0s (391 kB/s)
Selecting previously unselected
(Reading database ... 423327 fil
Preparing to unpack .../l3afpad_
Unpacking l3afpad (0.8.18.1.11-4-
Setting up l3afpad (0.8.18.1.11-
Processing triggers for desktop-
Processing triggers for hicolor-
Processing triggers for man-db (
Processing triggers for mailcap
Processing triggers for kali-menu
┌──(kali㉿kali)-[~]
└─$ sudo l3afpad
(Untitled)
File Edit Search Options Help
Right Ctrl
ENG
8:12 PM
7/20/2026

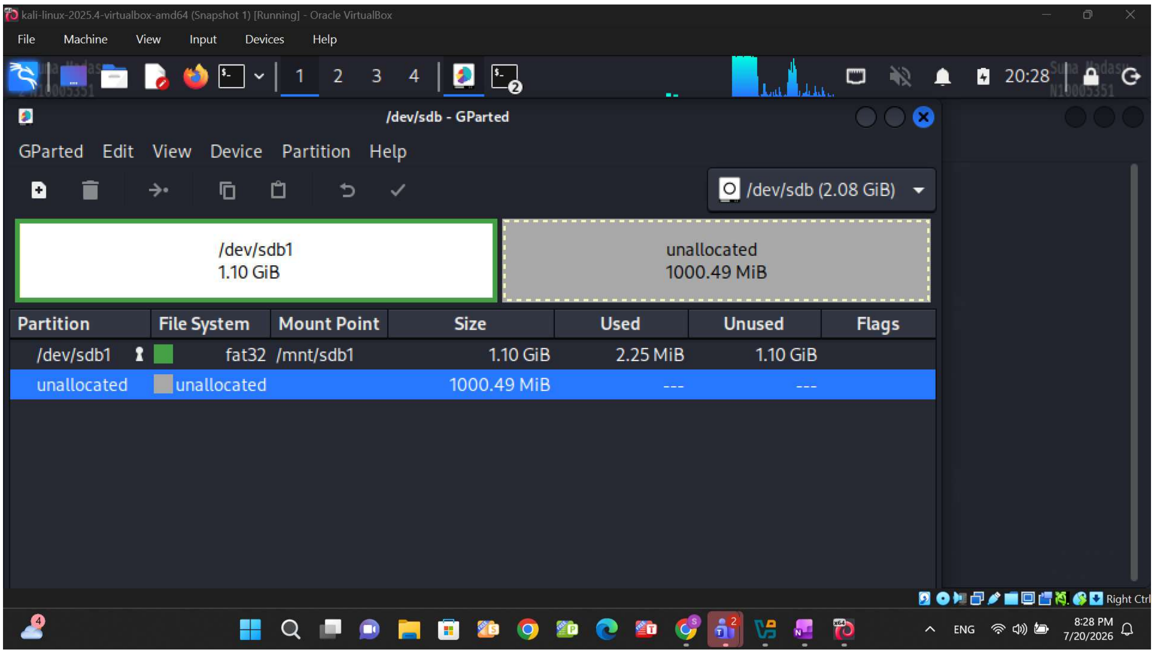

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Open
Recent
Home
Desktop
sf_VMShare
kali
Other Locations
etc
Name Size Type Modified
alternatives 2 Dec 2025
apache2 2 Dec 2025
apparmor 2 Dec 2025
apparmor.d 2 Dec 2025
apt 2 Dec 2025
arp-scan 2 Dec 2025
avahi 2 Dec 2025
bash_completion.d 2 Dec 2025
binfmt.d 17 Nov 2025
Character Coding:
Cancel
Open

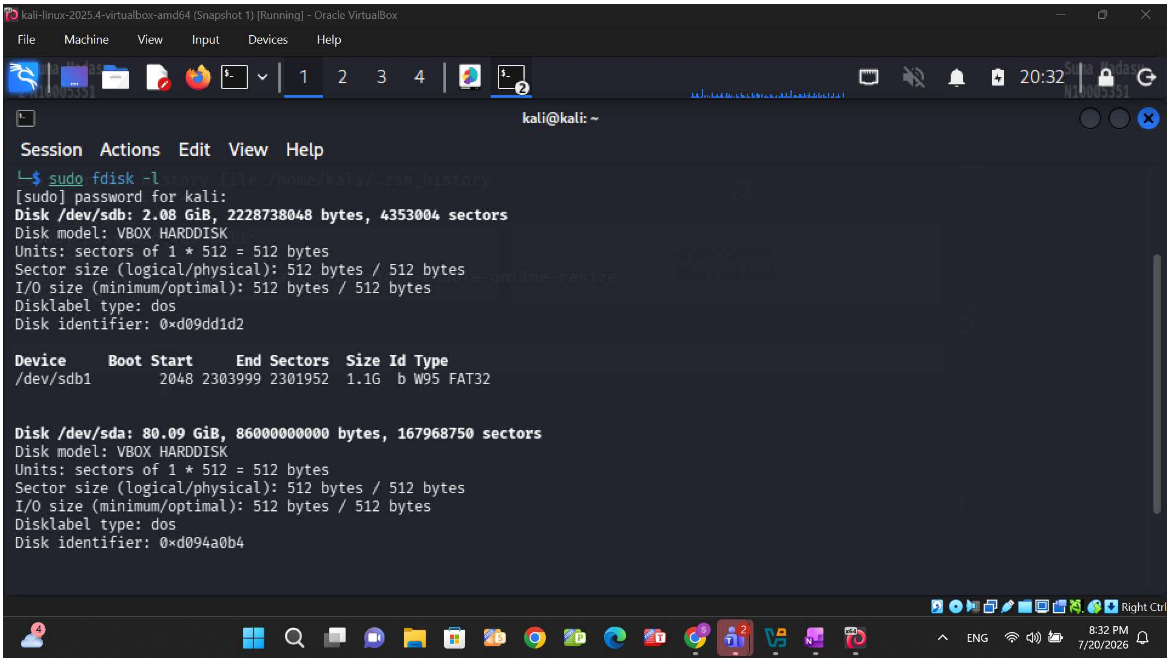

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Open
Recent
Home
Desktop
sf_VMShare
kali
Other Locations
etc
Name Size Type Modified
ethertypes 1.9 kB Text 15 Mar 2025
foremost.conf 9.1 kB Text 19 Sep 2022
fstab 139 bytes Text 2 Dec 2025
fuse.conf 725 bytes Text 20 Aug 2025
gai.conf 2.6 kB Text 28 Jan 2025
gprofng.rc 4.0 kB Text 22 Oct 2025
group 1.3 kB Text 6 Mar
group- 1.3 kB Text 6 Mar
gshadow 1.1 kB Text 6 Mar
Character Coding:
Cancel
Open

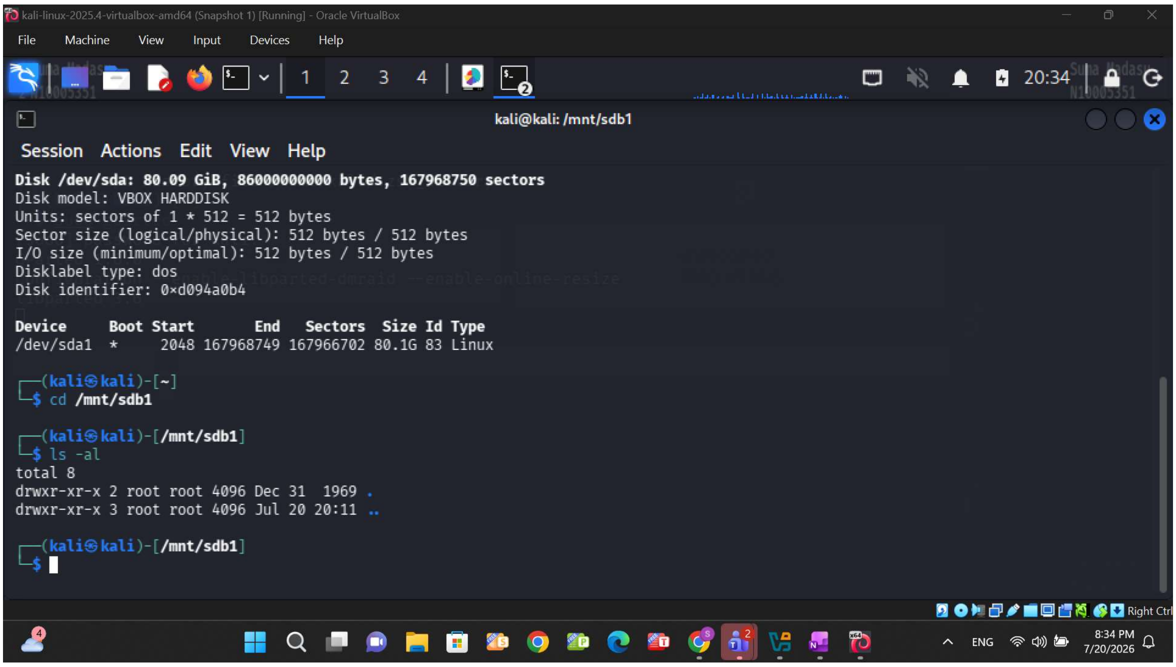

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
fstab
File Edit Search Options Help
UUID=8a84b405-db00-4b28-a5bc-48ecaa65880d / ext4
options> <dump> <pass>
/swap none swap defaults 0 0
Session Actions Edit View
Summary:
Upgrading: 0, Installing: 1,
Download size: 148 kB
Space needed: 505 kB / 58.7 GB
Fetched 148 kB in 0s (391 kB/s)
(kali㉿kali)-[~]
$ sudo l3afpad

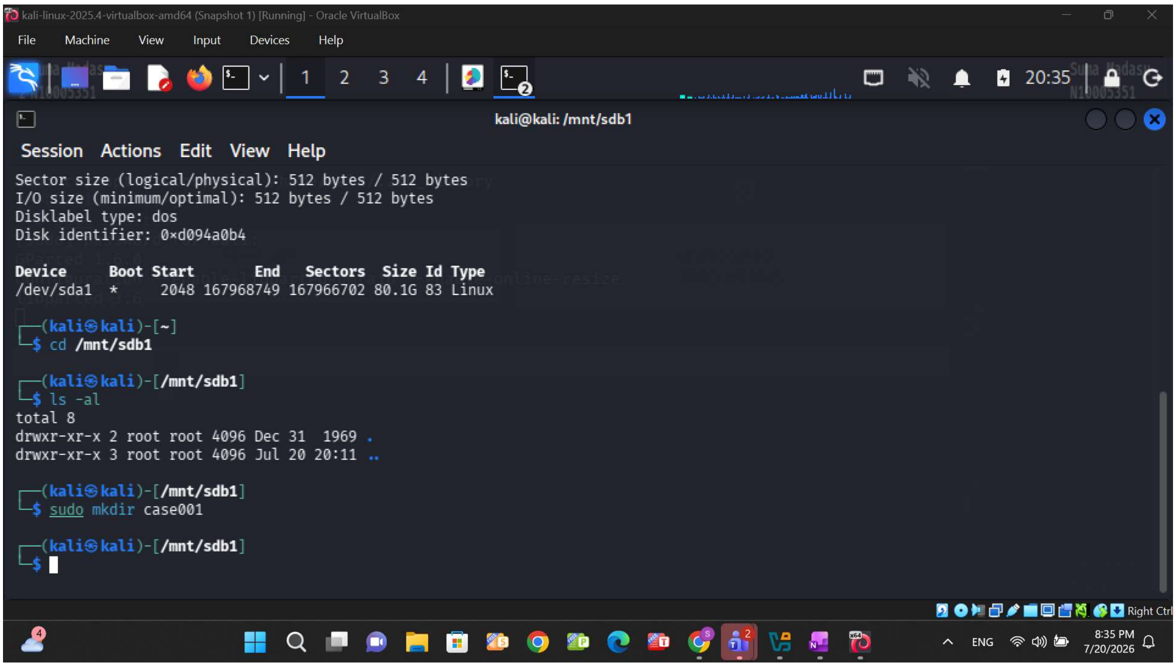

kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
20:23
fstab
Session Actions Edit View
File Edit Search Options Help
UUID=8a84b405-db00-4b28-a5bc-48ecaa65880d / ext4
options> <dump> <pass>
/swap none swap defaults 0 0
/dev/sdb1 /mnt/sdb1 vfat defaults 0 0
Summary:
Upgrading: 0, Installing: 1, R
Download size: 148 kB
Space needed: 505 kB / 58.7 GB
Get:1 http://http.kali.org/kali
Fetched 148 kB in 0s (391 kB/s)
Selecting previously unselected
(Reading database ... 423327 fil
Preparing to unpack .../l3afpad_
Unpacking l3afpad (0.8.18.1.11-4
Setting up l3afpad (0.8.18.1.11-
Processing triggers for desktop-
Processing triggers for hicolor-
Processing triggers for man-db (
Processing triggers for mailcap
Processing triggers for kali-men
(kali㉿kali)-[~]
$ sudo l3afpad
ENG
8:23 PM
7/20/2026


kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
20:24
*fstab
Session Actions Edit View
File Edit Search Options Help
UUID=8a84b405-db00-4b28-a5bc-48ecaa65880d / ext4
options> <dump> <pass>
/swap none swap defaults 0 0
/dev/sdb1 /mnt/sdb1 vfat defaults 0
Summary:
Upgrading: 0, Installing: 1, R
Download size: 148 kB
Space needed: 505 kB / 58.7 GB
Get:1 http://http.kali.org/kali
Fetched 148 kB in 0s (391 kB/s)
Selecting previously unselected
(Reading database ... 423327 fil
Preparing to unpack .../l3afpad_
Unpacking l3afpad (0.8.18.1.11-4
Setting up l3afpad (0.8.18.1.11-
Processing triggers for desktop-
Processing triggers for hicolor-
Processing triggers for man-db (
Processing triggers for mailcap
Processing triggers for kali-men
(kali㉿kali)-[~]
$ sudo l3afpad
ENG
8:24 PM
7/20/2026


kali-linux-2025.4-virtualbox-amd64 (Snapshot 1) [Running] - Oracle VirtualBox
/dev/sdb - GParted
GParted Edit View Device Partition Help
/dev/sdb (2.08 GiB)
/dev/sdb1
1.10 GiB
unallocated
1000.49 MiB
Partition File System Mount Point Size Used Unused Flags
/dev/sdb1 fat32 /mnt/sdb1 1.10 GiB 2.25 MiB 1.10 GiB
unallocated unallocated 1000.49 MiB --- ---


kali@kali: ~
Session Actions Edit View Help
└─$ sudo fdisk -l
[sudo] password for kali:
Disk /dev/sdb: 2.08 GiB, 2228738048 bytes, 4353004 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd09dd1d2
Device Boot Start End Sectors Size Id Type
/dev/sdb1 2048 2303999 2301952 1.1G b W95 FAT32
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd094a0b4


kali@kali: /mnt/sdb1
Session Actions Edit View Help
Disk /dev/sda: 80.09 GiB, 86000000000 bytes, 167968750 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xd094a0b4
Device Boot Start End Sectors Size Id Type
/dev/sda1 * 2048 167968749 167966702 80.1G 83 Linux
┌──(kali㉿kali)-[~]
└─$ cd /mnt/sdb1
┌──(kali㉿kali)-[/mnt/sdb1]
└─$ ls -al
total 8
drwxr-xr-x 2 root root 4096 Dec 31 1969 .
drwxr-xr-x 3 root root 4096 Jul 20 20:11 ..
┌──(kali㉿kali)-[/mnt/sdb1]
└─$



4. Key Forensic Learnings and Principles

Executing these procedures reinforced several foundational concepts necessary for operating as a security analyst and maintaining evidentiary integrity:

- **Acquisition Methodologies:** Forensic data can be systematically acquired via disk-to-image files, disk-to-disk copies, logical acquisitions, or sparse data copies of distinct files and folders.
- **Data Compression Standards:** Forensic imaging must rely exclusively on lossless compression, which preserves the original data exactly upon restoration, whereas lossy compression permanently alters data structures.
- **Validation Protocols:** It is a strict requirement to validate all acquired data to prove it has not been modified or tampered with. This validation is mathematically achieved using built-in forensic tool features, hexadecimal editors, or Linux hashing utilities such as md5sum and sha1sum.
- **Contingency Operations:** To prevent single points of failure, analysts should secure two separate acquisitions—one compressed and one uncompressed—if target storage capacity permits.
- **Write-Blocking Defenses:** When capturing data from suspect drives, physical write-blocking devices or software utilities must be deployed to prevent accidental data overwrites. Notably, confusing the input (if=) and output (of=) variables within a dd command can write data to the wrong drive, destroying digital evidence.
- **Optimized Tooling:** While the standard Linux dd utility is functional, dcfldd is structurally optimized for digital forensics. It is the preferred Linux alternative due to its integrated hashing and verification capabilities.